

1. What does AI stand for?

- A) Artificial Intelligence
- B) Advanced Interface
- C) Automated Integration
- D) Artificial Integration

Answer: A) Artificial Intelligence

2. Which of the following is an example of narrow AI?

- A) A self-learning robot
- B) A voice assistant like Siri
- C) A superintelligent AI
- D) A self-aware AI

Answer: B) A voice assistant like Siri

3. What is machine learning in the context of AI?

- A) AI that mimics human emotions
- B) The ability of machines to learn from data and improve performance over time
- C) AI that understands natural language
- D) AI that solves puzzles

Answer: B) The ability of machines to learn from data and improve performance over time

4. Which of these is a subfield of machine learning?

- A) Data Mining
- B) Natural Language Processing
- C) Neural Networks
- D) All of the above

Answer: D) All of the above

5. What is a neural network in AI?

- A) A network that connects various devices together
- B) A computing system inspired by the structure of the human brain
- C) A database that stores information
- D) A method of machine learning based on logic

Answer: B) A computing system inspired by the structure of the human brain

6. What is the main purpose of Natural Language Processing (NLP)?

- A) To help AI recognize images
- B) To allow AI to process and understand human language
- C) To improve machine learning models
- D) To make AI self-aware

Answer: B) To allow AI to process and understand human language

7. What is an example of AI used in healthcare?

- A) A self-driving car
- B) A chatbot for customer service
- C) AI for diagnosing diseases from medical images
- D) A recommendation system for shopping

Answer: C) AI for diagnosing diseases from medical images

8. Which of the following is a challenge faced by AI?

- A) Ability to handle large amounts of data
- B) AI's inability to understand complex human emotions
- C) Lack of hardware for computation
- D) All of the above

Answer: D) All of the above

9. What does NLP stand for in the context of AI?

- A) Neural Language Processing
- B) Natural Language Processing
- C) Non-Linguistic Programming
- D) Neural Learning Process

Answer: B) Natural Language Processing

10. Which of the following is a common task in NLP?

- A) Image recognition
- B) Sentiment analysis
- C) Pathfinding
- D) Game playing

Answer: B) Sentiment analysis

11. What is the main goal of Natural Language Processing (NLP)?

- A) To allow machines to process human language in a way that is meaningful
- B) To translate languages into numbers
- C) To create human-like emotions in machines
- D) To recognize speech without errors

Answer: A) To allow machines to process human language in a way that is meaningful

12. Which of these is an example of an NLP technique?

- A) Text classification
- B) Speech synthesis
- C) Image segmentation
- D) Pathfinding algorithms

Answer: A) Text classification

13. What does tokenization in NLP refer to?

- A) Breaking a text into smaller chunks like words or sentences
- B) Transforming text into numerical values
- C) Creating new words from existing ones
- D) Reversing the order of text

Answer: A) Breaking a text into smaller chunks like words or sentences

14. Which of the following is a challenge in NLP?

- A) Understanding ambiguity in human language
- B) Generating human-like images
- C) Solving mathematical equations
- D) Recognizing objects in images

Answer: A) Understanding ambiguity in human language

15. Which algorithm is commonly used for text classification in NLP?

- A) Decision Trees
- B) Naive Bayes
- C) K-Means Clustering
- D) Genetic Algorithms

Answer: B) Naive Bayes

16. What is the role of Named Entity Recognition (NER) in NLP?

- A) Identifying the language of a text
- B) Recognizing specific entities such as names of people, organizations, and locations
- C) Translating text into different languages
- D) Generating new text based on given prompts

Answer: B) Recognizing specific entities such as names of people, organizations, and locations

17. What is NLTK in Python?

- A) A library for image processing
- B) A toolkit for working with natural language data
- C) A framework for web development
- D) A package for data visualization

Answer: B) A toolkit for working with natural language data

18. Which of the following is a primary function of NLTK?

- A) Data mining
- B) Natural language processing tasks like tokenization and part-of-speech tagging
- C) Machine learning model building
- D) Data visualization

Answer: B) Natural language processing tasks like tokenization and part-of-speech tagging

19. How do you install the NLTK package in Python?

- A) pip install nltk
- B) install nltk
- C) pip get nltk
- D) python install nltk

Answer: A) pip install nltk

20. What is the function `nltk.word_tokenize()` used for?

- A) To remove stopwords from a text
- B) To convert text into lowercase
- C) To split text into words
- D) To perform part-of-speech tagging

Answer: C) To split text into words

21. Which of the following is a function in NLTK used to tag parts of speech?

- A) `nltk.pos_tag()`
- B) `nltk.tokenize()`
- C) `nltk.stem()`
- D) `nltk.tag()`

Answer: A) `nltk.pos_tag()`

22. What does the function `nltk.FreqDist()` in NLTK do?

- A) It creates a list of frequently occurring words in a text
- B) It generates a random text
- C) It converts text into numerical values
- D) It removes punctuation from a text

Answer: A) It creates a list of frequently occurring words in a text

23. Which corpus is included in NLTK for analyzing English text?

- A) Reuters
- B) Brown
- C) Gutenberg
- D) All of the above

Answer: D) All of the above

24. What is the purpose of the NLTK function `nltk.stem.WordNetLemmatizer()`?

- A) To count the frequency of words
- B) To tokenize text
- C) To remove stopwords from a sentence
- D) To reduce words to their base or root form

Answer: D) To reduce words to their base or root form

25. What is a time series data?

- A) Data that is ordered chronologically
- B) Data collected in intervals from the past
- C) Data that changes over time
- D) Data with only numeric values

Answer: A) Data that is ordered chronologically

26. Which of the following is a key characteristic of time series data?

- A) The data points are independent of each other
- B) Data is collected at different time intervals
- C) The data is ordered by time
- D) Time series data cannot have trends or seasonality

Answer: C) The data is ordered by time

27. Which of the following is an example of time series data?

- A) Stock market prices recorded daily
- B) Sales of a product across different regions
- C) Heights of individuals in a group
- D) Temperature of various cities at one point in time

Answer: A) Stock market prices recorded daily

28. What does trend in a time series data refer to?

- A) The cyclical patterns over time
- B) The long-term movement in the data
- C) Random fluctuations in the data
- D) Seasonal patterns occurring every year

Answer: B) The long-term movement in the data

29. What is seasonality in time series data?

- A) Random noise in data
- B) Repeating patterns or cycles over a fixed period, like monthly or yearly
- C) A sudden spike in data
- D) The increase or decrease in data over time

Answer: B) Repeating patterns or cycles over a fixed period, like monthly or yearly

30. Which of the following is used to make predictions on time series data?

- A) Linear regression
- B) Random forest
- C) Autoregressive Integrated Moving Average (ARIMA)
- D) Decision Trees

Answer: C) Autoregressive Integrated Moving Average (ARIMA)

31. What is the main purpose of decomposing time series data?

- A) To convert the data into a linear form
- B) To break the data into trend, seasonality, and residual components
- C) To convert categorical data into numerical values
- D) To eliminate missing values

Answer: B) To break the data into trend, seasonality, and residual components

32. What does autocorrelation in time series data refer to?

- A) The correlation of the data with itself at different time lags
- B) The correlation between two different datasets
- C) The difference between actual and predicted values
- D) The seasonal variation in the data

Answer: A) The correlation of the data with itself at different time lags

33. What is speech recognition?

- A) The ability to recognize faces from audio signals
- B) The process of converting audio signals into text

- C) The ability to recognize speech patterns in written text
- D) The process of translating text into speech

Answer: B) The process of converting audio signals into text

34. Which of the following is a popular speech recognition API?

- A) TensorFlow
- B) Google Speech-to-Text
- C) OpenCV
- D) PyTorch

Answer: B) Google Speech-to-Text

35. What does "Automatic Speech Recognition" (ASR) refer to?

- A) A method of analyzing the meaning behind speech
- B) A system that converts speech into phonetic symbols
- C) The technology that enables machines to transcribe spoken language into written text
- D) A system that generates human-like speech

Answer: C) The technology that enables machines to transcribe spoken language into written text

36. Which of the following challenges is commonly faced by speech recognition systems?

- A) Understanding background noise
- B) Recognizing multiple speakers
- C) Accurately transcribing homophones
- D) All of the above

Answer: D) All of the above

37. In speech recognition, what does "phoneme" refer to?

- A) A single word in a sentence
- B) A unit of sound that distinguishes one word from another
- C) A complete sentence of speech
- D) A type of noise in speech

Answer: B) A unit of sound that distinguishes one word from another

38. Which machine learning model is most commonly used for speech recognition tasks?

- A) Decision Trees
- B) Convolutional Neural Networks (CNNs)
- C) Recurrent Neural Networks (RNNs)
- D) Support Vector Machines (SVMs)

Answer: C) Recurrent Neural Networks (RNNs)

39. What is the role of "language models" in speech recognition?

- A) They improve the speed of recognition
- B) They help in understanding the meaning of the words
- C) They predict the probability of word sequences in spoken language
- D) They convert speech into other languages

Answer: C) They predict the probability of word sequences in spoken language

40. Which of the following is a typical application of speech recognition technology?

- A) Voice-activated assistants like Siri or Alexa
- B) Image classification
- C) Text summarization
- D) Sentiment analysis

Answer: A) Voice-activated assistants like Siri or Alexa

41. What is a heuristic search?

- A) A search algorithm that guarantees the optimal solution
- B) A search technique that uses a heuristic function to guide the search process
- C) A random search method
- D) A search method based on brute-force exploration

Answer: B) A search technique that uses a heuristic function to guide the search process

42. In a heuristic search, what is the purpose of the heuristic function?

- A) To explore the entire search space
- B) To calculate the cost of the solution
- C) To estimate the "cost" of reaching the goal from a given state
- D) To reduce the size of the search tree

Answer: C) To estimate the "cost" of reaching the goal from a given state

43. Which of the following search algorithms uses a heuristic function?

- A) Depth-First Search (DFS)
- B) Breadth-First Search (BFS)
- C) A* Search
- D) Uniform Cost Search

Answer: C) A* Search

44. What is the main advantage of using heuristic search algorithms?

- A) They guarantee finding the shortest path
- B) They are always faster than non-heuristic searches
- C) They can provide a good solution quickly, even in large search spaces
- D) They never miss the optimal solution

Answer: C) They can provide a good solution quickly, even in large search spaces

45. Which of the following is a typical example of a heuristic function in search problems?

- A) The exact cost to reach the goal
- B) The number of steps already taken
- C) The straight-line distance (or Euclidean distance) from the current state to the goal
- D) The depth of the current search tree

Answer: C) The straight-line distance (or Euclidean distance) from the current state to the goal

46. What is the major drawback of using greedy best-first search?

- A) It always finds the optimal solution
- B) It does not guarantee the shortest path to the goal
- C) It requires less memory than A*
- D) It explores all possible solutions

Answer: B) It does not guarantee the shortest path to the goal

*47. What does the A search algorithm combine?**

- A) Depth-First Search and Breadth-First Search
- B) The heuristic function and the cost of the path to the goal
- C) The greedy best-first search with depth-limited search
- D) A brute-force search and a random search

Answer: B) The heuristic function and the cost of the path to the goal

48. In the context of heuristic search, what does the term "admissibility" refer to?

- A) The search algorithm guarantees finding the goal
- B) The heuristic function never overestimates the cost to reach the goal
- C) The algorithm will always find the shortest path
- D) The search space is finite and bounded

Answer: B) The heuristic function never overestimates the cost to reach the goal

49. In the context of AI, what is the primary objective in game-playing algorithms?

- A) To create graphics for the game
- B) To maximize the player's score
- C) To make the AI play the game as efficiently as possible
- D) To simulate human-like behavior in the game

Answer: C) To make the AI play the game as efficiently as possible

50. Which of the following game-playing algorithms is most commonly used in two-player zero-sum games?

- A) Genetic Algorithm
- B) Minimax Algorithm
- C) Naive Bayes
- D) K-Means Clustering

Answer: B) Minimax Algorithm

51. What does the "alpha-beta pruning" technique optimize in game-playing AI?

- A) The decision-making process by eliminating unnecessary branches in the search tree
- B) The graphical rendering of the game

- C) The number of possible moves
- D) The memory usage of the AI algorithm

Answer: A) The decision-making process by eliminating unnecessary branches in the search tree

52. In the context of AI in games, what is a "heuristic function"?

- A) A function that calculates the best move by evaluating the possible outcomes
- B) A random number generator for choosing moves
- C) A function to make the AI act unpredictably
- D) A function that simulates a player's emotions

Answer: A) A function that calculates the best move by evaluating the possible outcomes

53. Which of the following is an example of a game that AI has successfully played using deep learning techniques?

- A) Chess
- B) Tic-Tac-Toe
- C) Go
- D) Snakes and Ladders

Answer: C) Go

54. What is the primary purpose of using reinforcement learning in gaming AI?

- A) To teach the AI how to play games through trial and error and improve performance over time
- B) To reduce the complexity of the game's algorithm
- C) To enable AI to simulate real-time actions in the game world
- D) To generate random strategies for the game

Answer: A) To teach the AI how to play games through trial and error and improve performance over time

55. In AI game-playing, what is "Monte Carlo Tree Search" (MCTS) used for?

- A) To evaluate a large number of possible game moves based on random simulations
- B) To prune the game tree for faster decision-making

- C) To generate heuristic values for each move
- D) To predict the player's next move based on historical data

Answer: A) To evaluate a large number of possible game moves based on random simulations

56. What is the main challenge when designing AI for real-time strategy (RTS) games?

- A) Finding the best game move
- B) Handling massive search spaces and long decision times
- C) Simulating human-like emotions in the AI
- D) Ensuring the AI plays in a visually pleasing way

Answer: B) Handling massive search spaces and long decision times

57. In reinforcement learning, what is the agent trying to maximize?

- A) The reward
- B) The environment
- C) The time step
- D) The exploration rate

Answer: A) The reward

58. What is the key difference between reinforcement learning and supervised learning?

- A) Reinforcement learning uses labeled data, while supervised learning does not
- B) Reinforcement learning learns from feedback from the environment, while supervised learning learns from labeled data
- C) Reinforcement learning is faster than supervised learning
- D) Reinforcement learning does not require a reward function

Answer: B) Reinforcement learning learns from feedback from the environment, while supervised learning learns from labeled data

59. What is a "reward" in reinforcement learning?

- A) The punishment for making the wrong decision
- B) The outcome or feedback received after taking an action in the environment
- C) The time taken to complete a task
- D) A random number generated by the agent

Answer: B) The outcome or feedback received after taking an action in the environment

60. What is the term "exploration" in the context of reinforcement learning?

- A) The process of finding the optimal policy
- B) Trying new actions in order to discover their effects
- C) Following the best-known strategy without any variation
- D) Memorizing the rewards received from actions

Answer: B) Trying new actions in order to discover their effects

61. What does the "discount factor" (γ) represent in reinforcement learning?

- A) The agent's learning rate
- B) The value of future rewards relative to immediate rewards
- C) The probability of the agent choosing a random action
- D) The reward given for each action

Answer: B) The value of future rewards relative to immediate rewards

62. Which algorithm is commonly used in reinforcement learning to estimate the action-value function?

- A) K-means
- B) Q-learning
- C) Linear regression
- D) Naive Bayes

Answer: B) Q-learning

63. What is the "policy" in reinforcement learning?

- A) The environment in which the agent operates
- B) The strategy used by the agent to determine the next action based on its state
- C) The reward function for the agent
- D) The set of actions available to the agent

Answer: B) The strategy used by the agent to determine the next action based on its state

64. What is the term for learning in reinforcement learning where an agent learns from interactions with the environment without any prior knowledge of the environment?

- A) Off-policy learning
- B) On-policy learning
- C) Model-based learning
- D) Unsupervised learning

Answer: B) On-policy learning

65. What is the primary inspiration for genetic algorithms?

- A) Natural language processing
- B) Evolution and natural selection in biological systems
- C) Neural networks
- D) Linear programming

Answer: B) Evolution and natural selection in biological systems

66. Which of the following operations is NOT part of the basic genetic algorithm?

- A) Selection
- B) Crossover
- C) Mutation
- D) Gradient descent

Answer: D) Gradient descent

67. What is the purpose of the "selection" process in a genetic algorithm?

- A) To generate a new population of solutions
- B) To randomly mutate the population
- C) To select the fittest individuals for reproduction
- D) To evaluate the fitness of each individual

Answer: C) To select the fittest individuals for reproduction

68. In genetic algorithms, what does "crossover" refer to?

- A) The process of copying the fitness of an individual into a new individual
- B) The merging of two parent solutions to create a new offspring
- C) The mutation of genes in a solution
- D) The selection of the fittest individual in the population

Answer: B) The merging of two parent solutions to create a new offspring

69. In the context of genetic algorithms, what is "mutation"?

- A) The elimination of weak individuals from the population
- B) The modification of a selected individual's genes to introduce diversity
- C) The recombination of two parents to create offspring
- D) The cloning of the best individual solution

Answer: B) The modification of a selected individual's genes to introduce diversity

70. What does the "fitness function" do in a genetic algorithm?

- A) It selects the parents for crossover
- B) It measures how well a solution solves the problem at hand
- C) It mutates the genes of an individual
- D) It calculates the cost of each individual solution

Answer: B) It measures how well a solution solves the problem at hand

71. In genetic algorithms, what is the "population"?

- A) The collection of all possible solutions
- B) The current set of candidate solutions in the algorithm
- C) The best solution found so far
- D) The initial random solutions generated by the algorithm

Answer: B) The current set of candidate solutions in the algorithm

72. What is one of the major advantages of genetic algorithms?

- A) They guarantee the optimal solution in all cases
- B) They can solve any problem without the need for a fitness function
- C) They are particularly useful for solving complex optimization problems with large search spaces
- D) They do not require randomness in the search process

Answer: C) They are particularly useful for solving complex optimization problems with large search spaces

73. What is the primary goal of computer vision?

- A) To generate graphics from scratch
- B) To enable computers to understand and interpret visual information from the world

- C) To analyze text in images
- D) To create realistic 3D models of objects

Answer: B) To enable computers to understand and interpret visual information from the world

74. Which of the following is a common application of computer vision?

- A) Handwriting recognition
- B) Text-to-speech conversion
- C) Voice recognition
- D) Natural language understanding

Answer: A) Handwriting recognition

75. In computer vision, what is an image segmentation task?

- A) Dividing an image into smaller patches for analysis
- B) Identifying the main object in an image
- C) Classifying objects within the image
- D) Dividing an image into meaningful regions or segments

Answer: D) Dividing an image into meaningful regions or segments

76. What is the role of convolutional neural networks (CNNs) in computer vision?

- A) To enhance the visual quality of images
- B) To generate 3D models from images
- C) To automatically learn features and patterns from images for classification and detection
- D) To segment images based on color patterns

Answer: C) To automatically learn features and patterns from images for classification and detection

77. What does the "feature extraction" process in computer vision involve?

- A) Creating 3D models of objects in an image
- B) Identifying the key characteristics or patterns in an image that can be used for further analysis
- C) Converting an image into a black-and-white version
- D) Reducing the resolution of an image for faster processing

Answer: B) Identifying the key characteristics or patterns in an image that can be used for further analysis

78. Which of the following is an example of an object detection task in computer vision?

- A) Detecting the edges of objects in an image
- B) Identifying the presence and location of a person in an image
- C) Recognizing the emotion expressed in a face
- D) Enhancing the brightness of an image

Answer: B) Identifying the presence and location of a person in an image

79. What is optical character recognition (OCR) used for in computer vision?

- A) To convert images of text into editable text
- B) To detect the presence of objects in images
- C) To analyze video content for motion detection
- D) To segment images into different regions

Answer: A) To convert images of text into editable text

80. What is the purpose of using data augmentation techniques in computer vision?

- A) To reduce the computational power required for image processing
- B) To artificially increase the size of the dataset by generating variations of the original images
- C) To enhance the visual quality of images
- D) To reduce the complexity of the neural network

Answer: B) To artificially increase the size of the dataset by generating variations of the original images

81. What is the main advantage of deep learning over traditional machine learning?

- A) It requires less data
- B) It can automatically extract features from raw data
- C) It is faster to train
- D) It always outperforms traditional methods

Answer: B) It can automatically extract features from raw data

82. Which layer in a neural network performs the task of combining the weighted inputs and applying an activation function?

- A) Input layer
- B) Output layer
- C) Hidden layer
- D) Convolutional layer

Answer: C) Hidden layer

83. What is the primary purpose of convolutional layers in a Convolutional Neural Network (CNN)?

- A) To handle sequential data
- B) To reduce the spatial dimensions of the input
- C) To apply filters to detect spatial hierarchies and features
- D) To perform classification tasks

Answer: C) To apply filters to detect spatial hierarchies and features

84. Which of the following activation functions is most commonly used in the hidden layers of deep learning models?

- A) Sigmoid
- B) Tanh
- C) ReLU (Rectified Linear Unit)
- D) Softmax

Answer: C) ReLU (Rectified Linear Unit)

85. What is a major benefit of using the ReLU activation function over the sigmoid or tanh functions?

- A) It guarantees faster training by preventing the vanishing gradient problem
- B) It ensures that the output is always positive
- C) It uses a smooth transition between input values
- D) It allows negative values to be passed through

Answer: A) It guarantees faster training by preventing the vanishing gradient problem

86. What is the role of the "pooling" layer in Convolutional Neural Networks (CNNs)?

- A) To reduce the number of parameters and computation by downsampling the feature maps
- B) To generate the final output of the network
- C) To increase the resolution of the image
- D) To perform feature extraction

Answer: A) To reduce the number of parameters and computation by downsampling the feature maps

87. In deep learning, what is "overfitting"?

- A) When the model performs well on both the training and test data
- B) When the model performs poorly on both the training and test data
- C) When the model performs well on training data but poorly on unseen data
- D) When the model is undertrained and does not capture patterns in the data

Answer: C) When the model performs well on training data but poorly on unseen data

88. What is "batch normalization" in deep learning used for?

- A) To optimize the learning rate
- B) To reduce overfitting
- C) To normalize the input features of the network during training
- D) To make the activation functions linear

Answer: C) To normalize the input features of the network during training

89. Which type of neural network is best suited for processing sequential data, such as time series or natural language?

- A) Convolutional Neural Networks (CNNs)
- B) Recurrent Neural Networks (RNNs)
- C) Multi-layer Perceptrons (MLPs)
- D) Generative Adversarial Networks (GANs)

Answer: B) Recurrent Neural Networks (RNNs)

90. What is the primary purpose of the softmax activation function in deep learning?

- A) To map the output to a binary decision
- B) To normalize the output into a probability distribution

- C) To ensure the output is always positive
- D) To prevent overfitting

Answer: B) To normalize the output into a probability distribution

91. What is a common method for dealing with overfitting in deep learning models?

- A) Increasing the number of parameters
- B) Decreasing the number of training examples
- C) Using dropout or regularization techniques
- D) Using a more complex model architecture

Answer: C) Using dropout or regularization techniques

92. Which of the following is an important component in the process of training a deep learning model?

- A) Hyperparameter tuning
- B) Feature engineering
- C) Neural architecture search
- D) All of the above

Answer: D) All of the above

93. In AI gaming, what is the purpose of the "minimax" algorithm?

- A) To select the most optimal action in multiplayer games
- B) To reduce the game tree's search space for faster decisions
- C) To find the best move by minimizing the maximum possible loss
- D) To generate random moves in complex games

Answer: C) To find the best move by minimizing the maximum possible loss

94. What is the primary function of "Monte Carlo Tree Search" (MCTS) in game AI?

- A) To simulate all possible moves and evaluate them using random simulations
- B) To identify the shortest path in a maze
- C) To estimate the time required to complete the game
- D) To detect cheating behavior in players

Answer: A) To simulate all possible moves and evaluate them using random simulations

95. Which deep learning architecture is typically used to build AI that can play video games directly from pixels (as seen in DeepMind's Atari game-playing AI)?

- A) Convolutional Neural Networks (CNNs)
- B) Recurrent Neural Networks (RNNs)
- C) Decision Trees
- D) Naive Bayes Classifier

Answer: A) Convolutional Neural Networks (CNNs)

96. In AI-driven games, what is "pathfinding"?

- A) Identifying the best strategies for players
- B) Determining the most efficient route for characters to move in a game environment
- C) Tracking the score of each player
- D) Generating realistic sound effects

Answer: B) Determining the most efficient route for characters to move in a game environment

97. What is "AI NPC" short for, and what is its role in gaming?

- A) Artificial Intelligence Network Protocol; to connect players in online games
- B) Artificial Intelligence Non-Player Character; to simulate intelligent behavior in characters controlled by the game
- C) Automated Interactive Non-Player Communication; to enhance player interaction
- D) None of the above

Answer: B) Artificial Intelligence Non-Player Character; to simulate intelligent behavior in characters controlled by the game

98. In the context of computer vision, what does the term "optical character recognition" (OCR) refer to?

- A) Detecting objects within an image
- B) Converting images of text into editable text
- C) Enhancing the quality of images
- D) Creating 3D models from images

Answer: B) Converting images of text into editable text

99. What is the purpose of using a "softmax" function in computer vision models?

- A) To convert the network output into a probability distribution for classification
- B) To enhance the resolution of an image
- C) To combine multiple images into one
- D) To detect edges in an image

Answer: A) To convert the network output into a probability distribution for classification

100. What is the advantage of using data augmentation in computer vision?

- A) It decreases the accuracy of the model
- B) It helps to artificially expand the size of the training dataset by generating modified versions of images
- C) It increases the training time significantly
- D) It improves the resolution of the images

Answer: B) It helps to artificially expand the size of the training dataset by generating modified versions of images

6. What is the result of this code?

```
python
```

```
Copy
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```
x = 10
```

```
y = 3
```

```
print(x // y)
```

- A) 3
- B) 3.33
- C) 0
- D) Error

Answer: A) 3

7. Which of the following is NOT a mutable data type in Python?

- A) List
- B) Dictionary

- C) Set
- D) Tuple

Answer: D) Tuple

8. Which of the following methods can be used to add an element to a dictionary in Python?

- A) insert()
- B) append()
- C) add()
- D) update()

Answer: D) update()

9. What will be the output of this code?

```
python
```

```
Copy
```

```
x = 10
```

```
y = 20
```

```
print(x == y)
```

- A) True
- B) False
- C) 10
- D) 20

Answer: B) False

10. What is the default value of the end parameter in the print() function in Python?

- A) "\n"
- B) ""
- C) None
- D) "\t"

Answer: A) "\n"

16. What does the break statement do in Python?

- A) Exits the current loop
- B) Skips the current iteration of the loop
- C) Continues the loop to the next iteration
- D) Exits the function

Answer: A) Exits the current loop

17. Which of the following statements is used to exit a function in Python?

- A) return
- B) exit
- C) break
- D) quit

Answer: A) return

18. Which of the following is the correct way to comment in Python?

- A) // This is a comment
- B) # This is a comment
- C) <!-- This is a comment -->
- D) /* This is a comment */

Answer: B) # This is a comment

19. Which of the following is used to install external packages in Python?

- A) install()
- B) pip install
- C) import
- D) setup.py

Answer: B) pip install

20. What will the following code output?

python

Copy

```
x = [10, 20, 30, 40]
```

```
y = [40, 30, 20, 10]
```

```
print(x == y)
```

A) True

B) False

C) Error

D) None

Answer: B) False